

AECCS Final Presentation

Automated GDPR compliance analysis and PET evaluation for cookie consent systems.

CS475 Privacy-Enhancing Technologies

Research Question

- Do cookie consent systems actually protect users from non-essential tracking?
- How often do banners rely on dark patterns?
- Which PETs reduce exposure most effectively when consent systems fail?

What Was Built

- Crawler with no-interaction, accept, and reject states.
- Tracker classification using filter lists and heuristics.
- Dark-pattern detection and GDPR-oriented compliance scoring.
- Browser PET evaluation, CMP comparison, and differential privacy reporting.
- Visualization and HTML reporting pipeline with provenance checks.

Data Collection

- 1,000 websites sampled across 12 categories.
- Three states per site: pre-consent, accept-all, reject-all.
- EU geolocation via VPN and Playwright-based automation with anti-detection behavior.

CS 475 Theory Bridge (Project Scope)

Privacy + Consent Theory (WEEK1-2)

- Privacy by design/default means protection should exist before user interaction.
- Valid consent must be informed, specific, and actively given.
- Legal banner text can diverge from real tracking behavior.

Project mapping: we test enforceability using pre-consent, accept-all, and reject-all measurements instead of trusting wording.

Course anchor: WEEK1 and 2-Privacy.pdf, p.56-60, 77-78

Anonymization PET Theory (Ch3)

- k-anonymity and l-diversity motivate anonymization, but linkage risks remain.
- Differential privacy bounds what changes when one record is added or removed.
- Noisy aggregate release preserves trends while lowering disclosure risk.

Project mapping: we release only DP-protected aggregate results (epsilon sweep, with epsilon = 1.0 as the balanced choice).

Course anchor: Ch3-PETs for Data Anonymization.pdf, p.21, 52, 104-105

Bridge summary: course theory defines what valid consent and safe publication require; AECCS operationalizes both with 3-state measurement and DP reporting.

Current Repository Status

Area	Status
Software pipeline	Implemented
Mock/demo dataset	Reproducible
Real study list	Prepared with 100 domains
Real crawl outputs	Completed (97/100 sites, 1 March 2026)
Final empirical results	Completed (100-site study with 7 PETs)

Real-Study Findings (100 Sites, 1 March 2026)

Therefore we measured tracker behavior directly across consent states and report only evidence-backed findings.

- 89.7% of sites set pre-consent trackers before any user interaction.
- 82.0% of sites exhibit at least one dark pattern (most common: missing reject button at 69%).
- Average compliance score is 33.1 / 100 (median 24.0); 67% of sites received grade F.
- Brave Shields is the strongest browser PET (14.7% average tracker reduction).
- OneTrust is the strongest CMP as a privacy tool (PET score 33.4).
- Best combination: Brave Shields + OneTrust (estimated 43.2% effectiveness).
- Recommended DP epsilon: 1.0 for balanced privacy/utility when publishing aggregate statistics.

Based on real crawl data from data/real/, run ID real-study-20260301-final.

Completed Actions

- Real crawl and PET evaluation completed under data/real/ (1 March 2026).
- All 11 figures and HTML compliance report regenerated from real processed data.
- Demo findings replaced with real-study results across all documents.